Table 11: Additional comparison with other baselines in long-term forecasting tasks. We set the forecasting horizons $H \in \{24, 36, 48, 60\}$ for ILI and $\{96, 192, 336, 720\}$ for the others. A lower value indicates better performance. **Red**: the best, Blue: the second best.

Methods		TIME-LLM		N-BEATS		N-HiTS		AutoARIMA		AutoTheta		AutoETS	
Metric		MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
ETTh1	96	**0.362**	**0.392**	0.496	0.475	0.392	0.407	0.933	0.635	1.266	0.758	1.264	0.756
	192	**0.398**	**0.418**	0.544	0.504	0.442	0.438	0.868	0.621	1.188	0.749	1.181	0.745
	336	**0.430**	**0.427**	0.592	0.533	0.497	0.471	0.964	0.663	1.310	0.799	1.292	0.792
	720	**0.442**	**0.457**	0.639	0.588	0.559	0.533	1.043	0.705	1.510	0.882	1.405	0.842
	Avg	**0.408**	**0.423**	0.568	0.525	0.473	0.462	0.952	0.656	1.319	0.797	1.286	0.784
ETTh2	96	**0.268**	**0.328**	0.384	0.431	0.321	0.368	0.390	0.417	0.461	0.430	0.444	0.403
	192	**0.329**	**0.375**	0.496	0.493	0.398	0.421	0.545	0.492	0.754	0.537	0.771	0.461
	336	**0.368**	**0.409**	0.585	0.542	0.453	0.459	0.697	0.562	1.355	0.683	1.526	0.522
	720	**0.372**	**0.420**	0.792	0.651	0.775	0.609	0.907	0.658	3.971	1.061	5.183	0.633
	Avg	**0.334**	**0.383**	0.564	0.529	0.487	0.464	0.635	0.532	1.635	0.678	1.981	0.505
ETTm1	96	**0.272**	**0.334**	0.393	0.412	0.327	0.368	1.091	0.661	1.211	0.704	1.519	0.768
	192	**0.310**	**0.358**	0.425	0.427	0.376	0.400	1.119	0.682	1.237	0.724	1.535	0.784
	336	**0.352**	**0.384**	0.464	0.454	0.407	0.423	1.125	0.698	1.231	0.735	1.472	0.782
	720	**0.383**	**0.411**	0.521	0.488	0.471	0.456	1.243	0.745	1.394	0.801	1.591	0.825
	Avg	**0.329**	**0.372**	0.451	0.445	0.395	0.412	1.145	0.697	1.268	0.741	1.529	0.790
ETTm2	96	**0.161**	**0.253**	0.204	0.302	0.188	0.273	0.435	0.375	0.245	0.316	0.359	0.333
	192	**0.219**	**0.293**	0.282	0.358	0.274	0.338	0.995	0.494	0.413	0.401	0.756	0.396
	336	**0.271**	**0.329**	0.378	0.425	0.384	0.406	2.324	0.648	0.790	0.528	1.747	0.467
	720	**0.352**	**0.379**	0.555	0.523	0.501	0.488	9.064	1.020	2.451	0.847	6.856	0.639
	Avg	**0.251**	**0.313**	0.355	0.402	0.337	0.376	3.205	0.634	0.975	0.523	2.430	0.459
Weather	96	**0.147**	**0.201**	0.185	0.244	0.160	0.222	0.255	0.273	0.279	0.266	0.331	0.277
	192	**0.189**	**0.234**	0.225	0.282	0.202	0.265	0.390	0.353	0.337	0.316	0.498	0.345
	336	0.262	**0.279**	0.274	0.323	**0.253**	0.303	0.775	0.457	0.472	0.385	0.898	0.423
	720	**0.304**	**0.316**	0.340	0.373	0.323	0.354	2.898	0.707	0.818	0.526	2.820	0.580
	Avg	**0.225**	**0.257**	0.256	0.306	0.235	0.286	1.080	0.448	0.477	0.373	1.137	0.406
Electricity	96	**0.131**	**0.224**	0.233	0.327	0.184	0.275	0.520	0.466	0.653	0.532	0.650	0.526
	192	**0.152**	**0.241**	0.246	0.340	0.190	0.282	0.581	0.499	0.713	0.561	0.704	0.549
	336	**0.160**	**0.248**	0.262	0.355	0.205	0.298	0.602	0.515	0.797	0.603	0.766	0.577
	720	**0.192**	**0.298**	0.296	0.383	0.239	0.330	0.685	0.558	1.023	0.688	0.901	0.628
	Avg	**0.158**	**0.252**	0.259	0.351	0.205	0.296	0.597	0.510	0.797	0.596	0.755	0.570
Traffic	96	**0.362**	**0.248**	0.608	0.447	0.410	0.329	1.068	0.694	3.207	1.219	3.254	1.221
	192	**0.374**	**0.247**	0.605	0.448	0.414	0.330	1.380	0.775	3.407	1.262	3.569	1.264
	336	**0.385**	**0.271**	0.618	0.454	0.428	0.337	1.448	0.790	3.473	1.274	3.971	1.275
	720	**0.430**	**0.288**	0.650	0.467	0.456	0.354	1.481	0.799	3.952	1.382	6.784	1.379
	Avg	**0.388**	**0.264**	0.620	0.454	0.427	0.338	1.344	0.765	3.510	1.284	4.395	1.285
ILI	24	**1.285**	**0.727**	6.809	1.870	2.675	1.080	4.909	1.329	5.991	1.510	4.869	1.315
	36	**1.404**	**0.814**	6.850	1.890	3.081	1.194	5.079	1.440	5.922	1.539	4.917	1.422
	48	**1.523**	**0.807**	6.788	1.876	2.973	1.176	4.276	1.339	4.637	1.329	3.966	1.301
	60	**1.531**	**0.854**	6.908	1.893	3.259	1.232	3.855	1.276	4.378	1.345	3.540	1.229
	Avg	**1.435**	**0.801**	6.839	1.882	2.997	1.171	4.530	1.346	5.232	1.431	4.323	1.317
1^{st}Count		**40**		0		1		0		0		0	

Table 12: Full short-term time series forecasting results. The forecasting horizons are in [6, 48] and the last three rows are weighted averaged from all datasets under different sampling intervals. A lower value indicates better performance. **Red**: the best, Blue: the second best.

Methods		TIME-LLM	GPT4TS	TimesNet	PatchTST	N-HiTS	N-BEATS	ETSformer	LightTS	DLinear	FEDformer	Stationary	Autoformer	Informer	Reformer
Yearly	SMAPE	**13.419**	15.11	15.378	13.477	13.422	13.487	18.009	14.247	16.965	14.021	13.717	13.974	14.727	16.169
	MASE	**3.005**	3.565	3.554	3.019	3.056	3.036	4.487	3.109	4.283	3.036	3.078	3.134	3.418	3.800
	OWA	**0.789**	0.911	0.918	0.792	0.795	0.795	1.115	0.827	1.058	0.811	0.807	0.822	0.881	0.973
Quarterly	SMAPE	**10.110**	10.597	10.465	10.38	10.185	10.564	13.376	11.364	12.145	11.1	10.958	11.338	11.360	13.313
	MASE	**1.178**	1.253	1.227	1.233	1.18	1.252	1.906	1.328	1.520	1.35	1.325	1.365	1.401	1.775
	OWA	**0.889**	0.938	0.923	0.921	0.893	0.936	1.302	1.000	1.106	0.996	0.981	1.012	1.027	1.252
Monthly	SMAPE	12.980	13.258	13.513	**12.959**	13.059	13.089	14.588	14.014	13.514	14.403	13.917	13.958	14.062	20.128
	MASE	**0.963**	1.003	1.039	0.97	1.013	0.996	1.368	1.053	1.037	1.147	1.097	1.103	1.141	2.614
	OWA	**0.903**	0.931	0.957	0.905	0.929	0.922	1.149	0.981	0.956	1.038	0.998	1.002	1.024	1.927
Others	SMAPE	4.795	6.124	6.913	4.952	**4.711**	6.599	7.267	15.880	6.709	7.148	6.302	5.485	24.460	32.491
	MASE	3.178	4.116	4.507	3.347	**3.054**	4.43	5.240	11.434	4.953	4.041	4.064	3.865	20.960	33.355
	OWA	1.006	1.259	1.438	1.049	**0.977**	1.393	1.591	3.474	1.487	1.389	1.304	1.187	5.879	8.679
Average	SMAPE	**11.983**	12.69	12.88	12.059	12.035	12.25	14.718	13.525	13.639	13.16	12.780	12.909	14.086	18.200
	MASE	**1.595**	1.808	1.836	1.623	1.625	1.698	2.408	2.111	2.095	1.775	1.756	1.771	2.718	4.223
	OWA	**0.859**	0.94	0.955	0.869	0.869	0.896	1.172	1.051	1.051	0.949	0.930	0.939	1.230	1.775

E.2 ZERO-SHOT FORECASTING

The full results of zero-shot forecasting are summarized in Tab. 16. TIME-LLM remarkably surpasses the six most competitive time series models in zero-shot adaptation. Overall, we observe over **23.5%** and **12.4%** MSE and MAE reductions across all baselines on average. Our improvements are consistently significant on those typical cross-domain scenarios (e.g., ETTh2 → ETTh1 and ETTm2 → ETTm1), over **20.8%** and **11.3%** on average w.r.t. MSE and MAE. Significantly, TIME-LLM exhibits superior performance gains in comparison to LLMTime (Gruver et al., 2023), which employs a similarly sized backbone LLM (7B) and is the latest effort in leveraging LLMs for

Table 13: Additional short-term time series forecasting results on M3 (Quarterly). The forecasting horizon is 8. A lower value indicates better performance. **Red**: the best, Blue: the second best.

Methods	TIME-LLM	GPT4TS	TimesNet	PatchTST	N-HiTS	N-BEATS	DLinear	FEDformer
SMAPE	11.171	14.453	**10.410**	12.380	12.616	18.640	15.028	12.927
MRAE	3.282	5.035	3.310	**2.401**	4.271	4.612	2.793	3.653
MAPE	0.151	0.203	**0.140**	0.154	0.168	0.247	0.196	0.174

zero-shot time series forecasting. We attribute this success to our reprogramming framework being better at activating the LLM's knowledge transfer and reasoning capabilities in a resource-efficient manner when performing time series tasks.

F ABLATION STUDY

The full ablation results are in Tab. 17. We additionally compare the model performance under reprogramming and fine-tuning (with QLoRA Dettmers et al. (2023)) protocols. Our results indicate a clear performance gain of our approach compared to the QLoRA variant (**A.5**) by **19%** in average.

G EFFICIENCY COMPARISON WITH MODEL FINE-TUNING

Setups. We compare the efficiency of model fine-tuning (with QLoRA Dettmers et al. (2023)) and our proposed model reprogramming in this section with two different backbones, that is, Llama in 1/4 capacity (first 8 Transformer layers) and full capacity. Here, we adhere to the long-term forecasting protocol on ETTh1 to forecast two different steps (that is, 96 and 336 in this case) ahead. For the evaluation metrics, we report the total number of trainable parameters (in million), GPU memory (in mebibyte), and running time (seconds per iteration).

Results. Our results are given in Tab. 18. We see that model reprogramming remarkably results in better efficiency compared to parameter-efficient fine-tuning (PEFT) with QLoRA on long-range

Table 14: Full few-shot learning results on 10% training data. We use the same protocol as in Tab. 1.

Methods		TIME-LLM		GPT4TS		DLinear		PatchTST		TimesNet		FEDformer		Autoformer		Stationary		ETSformer		LightTS		Informer		Reformer	
Metric		MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
ETTh1	96	**0.448**	0.460	0.458	**0.456**	0.492	0.495	0.516	0.485	0.861	0.628	0.512	0.499	0.613	0.552	0.918	0.639	1.112	0.806	1.298	0.838	1.179	0.792	1.184	0.790
	192	**0.484**	**0.483**	0.570	0.516	0.565	0.538	0.598	0.524	0.797	0.593	0.624	0.555	0.722	0.598	0.915	0.629	1.155	0.823	1.322	0.854	1.199	0.806	1.295	0.850
	336	**0.589**	0.540	0.608	**0.535**	0.721	0.622	0.657	0.550	0.941	0.648	0.691	0.574	0.750	0.619	0.939	0.644	1.179	0.832	1.347	0.870	1.202	0.811	1.294	0.854
	720	**0.700**	**0.604**	0.725	0.591	0.986	0.743	0.762	0.610	0.877	0.641	0.728	0.614	0.721	0.616	0.887	0.645	1.273	0.874	1.534	0.947	1.217	0.825	1.223	0.838
	Avg	**0.556**	**0.522**	0.590	0.525	0.691	0.600	0.633	0.542	0.869	0.628	0.639	0.561	0.702	0.596	0.915	0.639	1.180	0.834	1.375	0.877	1.199	0.809	1.249	0.833
ETTh2	96	**0.275**	**0.326**	0.331	0.374	0.357	0.411	0.353	0.389	0.378	0.409	0.382	0.416	0.413	0.451	0.389	0.411	0.678	0.619	2.022	1.006	3.837	1.508	3.788	1.533
	192	**0.374**	**0.373**	0.402	0.411	0.569	0.519	0.403	0.414	0.490	0.467	0.478	0.474	0.474	0.477	0.473	0.455	0.785	0.666	2.329	1.104	3.856	1.513	3.552	1.483
	336	**0.406**	**0.429**	**0.406**	0.433	0.671	0.572	0.426	0.441	0.537	0.494	0.504	0.501	0.547	0.543	0.507	0.480	0.839	0.694	2.453	1.122	3.952	1.526	3.395	1.526
	720	**0.427**	**0.449**	0.449	0.464	0.824	0.648	0.477	0.480	0.510	0.491	0.499	0.509	0.516	0.523	0.477	0.472	1.273	0.874	3.816	1.407	3.842	1.503	3.205	1.401
	Avg	**0.370**	**0.394**	0.397	0.421	0.605	0.538	0.415	0.431	0.479	0.465	0.466	0.475	0.488	0.499	0.462	0.455	0.894	0.713	2.655	1.160	3.872	1.513	3.485	1.486
ETTm1	96	**0.346**	**0.388**	0.390	0.404	0.352	0.392	0.410	0.419	0.583	0.501	0.578	0.518	0.774	0.614	0.761	0.568	0.911	0.688	0.921	0.682	1.162	0.785	1.442	0.847
	192	**0.373**	0.416	0.429	0.423	0.382	**0.412**	0.437	0.434	0.630	0.528	0.617	0.546	0.754	0.592	0.781	0.574	0.955	0.703	0.957	0.701	1.172	0.793	1.444	0.862
	336	**0.413**	**0.426**	0.469	0.439	0.419	0.434	0.476	0.454	0.725	0.568	0.998	0.775	0.869	0.677	0.803	0.587	0.991	0.719	0.998	0.716	1.227	0.908	1.450	0.866
	720	**0.485**	0.476	0.569	0.498	0.490	**0.477**	0.681	0.556	0.769	0.549	0.693	0.579	0.810	0.630	0.844	0.581	1.062	0.747	1.007	0.719	1.207	0.797	1.366	0.850
	Avg	**0.404**	**0.427**	0.464	0.441	0.411	0.429	0.501	0.466	0.677	0.537	0.722	0.605	0.802	0.628	0.797	0.578	0.980	0.714	0.971	0.705	1.192	0.821	1.426	0.856
ETTm2	96	**0.177**	**0.261**	0.188	0.269	0.213	0.303	0.191	0.274	0.212	0.285	0.291	0.399	0.352	0.454	0.229	0.308	0.331	0.430	0.813	0.688	3.203	1.407	4.195	1.628
	192	**0.241**	**0.314**	0.251	0.309	0.278	0.345	0.252	0.317	0.270	0.323	0.307	0.379	0.694	0.691	0.291	0.343	0.400	0.464	1.008	0.768	3.112	1.387	4.042	1.601
	336	**0.274**	**0.327**	0.307	0.346	0.338	0.385	0.306	0.353	0.323	0.353	0.543	0.559	2.408	1.407	0.348	0.376	0.469	0.498	1.031	0.775	3.255	1.421	3.963	1.585
	720	**0.417**	**0.390**	0.426	0.417	0.436	0.440	0.433	0.427	0.474	0.449	0.712	0.614	1.913	1.166	0.461	0.438	0.589	0.557	1.096	0.791	3.909	1.543	3.711	1.532
	Avg	**0.277**	**0.323**	0.293	0.335	0.316	0.368	0.296	0.343	0.320	0.353	0.463	0.488	1.342	0.930	0.332	0.366	0.447	0.487	0.987	0.756	3.370	1.440	3.978	1.587
Weather	96	**0.161**	**0.210**	0.163	0.215	0.171	0.224	0.165	0.215	0.184	0.230	0.188	0.253	0.221	0.297	0.192	0.234	0.199	0.272	0.217	0.269	0.374	0.401	0.335	0.380
	192	**0.204**	**0.248**	0.210	0.254	0.215	0.263	0.210	0.257	0.245	0.283	0.250	0.304	0.270	0.322	0.269	0.295	0.279	0.332	0.259	0.304	0.552	0.478	0.522	0.462
	336	0.261	0.302	**0.256**	**0.292**	0.258	0.299	0.259	0.297	0.305	0.321	0.312	0.346	0.320	0.351	0.370	0.357	0.356	0.386	0.303	0.334	724	0.541	0.715	0.535
	720	**0.309**	**0.332**	0.321	0.339	0.320	0.346	0.332	0.346	0.381	0.371	0.387	0.393	0.390	0.396	0.441	0.405	0.437	0.448	0.377	0.382	0.739	0.558	0.611	0.500
	Avg	**0.234**	**0.273**	0.238	0.275	0.241	0.283	0.242	0.279	0.279	0.301	0.284	0.324	0.300	0.342	0.318	0.323	0.318	0.360	0.289	0.322	0.597	0.495	0.546	0.469
Electricity	96	**0.139**	0.241	**0.139**	**0.237**	0.150	0.253	0.140	0.238	0.299	0.373	0.231	0.323	0.261	0.348	0.420	0.466	0.599	0.587	0.350	0.425	1.259	0.919	0.993	0.784
	192	**0.151**	**0.248**	0.156	0.252	0.164	0.264	0.160	0.255	0.305	0.379	0.261	0.356	0.338	0.406	0.411	0.459	0.620	0.598	0.376	0.448	1.160	0.873	0.938	0.753
	336	**0.169**	**0.270**	0.175	**0.270**	0.181	0.282	0.180	0.276	0.319	0.391	0.360	0.445	0.410	0.474	0.434	0.473	0.662	0.619	0.428	0.485	1.157	0.872	0.925	0.745
	720	0.240	0.322	0.233	**0.317**	**0.223**	0.321	0.241	0.323	0.369	0.426	0.530	0.585	0.715	0.685	0.510	0.521	0.757	0.664	0.611	0.597	1.203	0.898	1.004	0.790
	Avg	**0.175**	0.270	0.176	**0.269**	0.180	0.280	0.180	0.273	0.323	0.392	0.346	0.427	0.431	0.478	0.444	0.480	0.660	0.617	0.441	0.489	1.195	0.891	0.965	0.768
Traffic	96	0.418	0.291	0.414	0.297	0.419	0.298	**0.403**	**0.289**	0.719	0.416	0.639	0.400	0.672	0.405	1.412	0.802	1.643	0.855	1.157	0.636	1.557	0.821	1.527	0.815
	192	**0.414**	**0.296**	0.426	0.301	0.434	0.305	0.415	**0.296**	0.748	0.428	0.637	0.416	0.727	0.424	1.419	0.806	1.641	0.854	1.207	0.661	1.454	0.765	1.538	0.817
	336	**0.421**	0.311	0.434	**0.303**	0.449	0.313	0.426	0.304	0.853	0.471	0.655	0.427	0.749	0.454	1.443	0.815	1.711	0.878	1.334	0.713	1.521	0.812	1.550	0.819
	720	**0.462**	**0.327**	0.487	0.337	0.484	0.336	0.474	0.331	1.485	0.825	0.722	0.456	0.847	0.499	1.539	0.837	2.660	1.157	1.292	0.726	1.605	0.846	1.588	0.833
	Avg	**0.429**	0.306	0.440	0.310	0.447	0.313	0.430	**0.305**	0.951	0.535	0.663	0.425	0.749	0.446	1.453	0.815	1.914	0.936	1.248	0.684	1.534	0.811	1.551	0.821
1st Count		**32**		9		3		3		0		0		0		0		0		0		0		0	

Table 15: Full few-shot learning results on 5% training data. We use the same protocol as in Tab. 1. '-' means that 5% time series is not sufficient to constitute a training set.

Methods		TIME-LLM		GPT4TS		DLinear		PatchTST		TimesNet		FEDformer		Autoformer		Stationary		ETSformer		LightTS		Informer		Reformer	
Metric		MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
$ETTh1$	96	**0.483**	**0.464**	0.543	0.506	0.547	0.503	0.557	0.519	0.892	0.625	0.593	0.529	0.681	0.570	0.952	0.650	1.169	0.832	1.483	0.91	1.225	0.812	1.198	0.795
	192	**0.629**	**0.540**	0.748	0.580	0.720	0.604	0.711	0.570	0.940	0.665	0.652	0.563	0.725	0.602	0.943	0.645	1.221	0.853	1.525	0.93	1.249	0.828	1.273	0.853
	336	0.768	0.626	0.754	0.595	0.984	0.727	0.816	0.619	0.945	0.653	**0.731**	**0.594**	0.761	0.624	0.935	0.644	1.179	0.832	1.347	0.87	1.202	0.811	1.254	0.857
	720	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
	Avg	**0.627**	**0.543**	0.681	0.560	0.750	0.611	0.694	0.569	0.925	0.647	0.658	0.562	0.722	0.598	0.943	0.646	1.189	0.839	1.451	0.903	1.225	0.817	1.241	0.835
$ETTh2$	96	**0.336**	**0.397**	0.376	0.421	0.442	0.456	0.401	0.421	0.409	0.420	0.390	0.424	0.428	0.468	0.408	0.423	0.678	0.619	2.022	1.006	3.837	1.508	3.753	1.518
	192	**0.406**	**0.425**	0.418	0.441	0.617	0.542	0.452	0.455	0.483	0.464	0.457	0.465	0.496	0.504	0.497	0.468	0.845	0.697	3.534	1.348	3.975	1.933	3.516	1.473
	336	**0.405**	**0.432**	0.408	0.439	1.424	0.849	0.464	0.469	0.499	0.479	0.477	0.483	0.486	0.496	0.507	0.481	0.905	0.727	4.063	1.451	3.956	1.520	3.312	1.427
	720	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
	Avg	**0.382**	**0.418**	0.400	0.433	0.694	0.577	0.827	0.615	0.439	0.448	0.463	0.454	0.441	0.457	0.470	0.489	0.809	0.681	3.206	1.268	3.922	1.653	3.527	1.472
$ETTm1$	96	**0.316**	0.377	0.386	0.405	0.332	**0.374**	0.399	0.414	0.606	0.518	0.628	0.544	0.726	0.578	0.823	0.587	1.031	0.747	1.048	0.733	1.130	0.775	1.234	0.798
	192	0.450	0.464	**0.440**	0.438	0.358	0.390	0.441	**0.436**	0.681	0.539	0.666	0.566	0.750	0.591	0.844	0.591	1.087	0.766	1.097	0.756	1.150	0.788	1.287	0.839
	336	0.450	0.424	0.485	0.459	**0.402**	**0.416**	0.499	0.467	0.786	0.597	0.807	0.628	0.851	0.659	0.870	0.603	1.138	0.787	1.147	0.775	1.198	0.809	1.288	0.842
	720	**0.483**	**0.471**	0.577	0.499	0.511	0.489	0.767	0.587	0.796	0.593	0.822	0.633	0.857	0.655	0.893	0.611	1.245	0.831	1.200	0.799	1.175	0.794	1.247	0.828
	Avg	0.425	0.434	0.472	0.450	**0.400**	**0.417**	0.526	0.476	0.717	0.561	0.730	0.592	0.796	0.620	0.857	0.598	1.125	0.782	1.123	0.765	1.163	0.791	1.264	0.826
$ETTm2$	96	**0.174**	**0.261**	0.199	0.280	0.236	0.326	0.206	0.288	0.220	0.299	0.229	0.320	0.232	0.322	0.238	0.316	0.404	0.485	1.108	0.772	3.599	1.478	3.883	1.545
	192	**0.215**	**0.287**	0.256	0.316	0.306	0.373	0.264	0.324	0.311	0.361	0.394	0.361	0.291	0.357	0.298	0.349	0.479	0.521	1.317	0.850	3.578	1.475	3.553	1.484
	336	**0.273**	**0.330**	0.318	0.353	0.380	0.423	0.334	0.367	0.338	0.366	0.378	0.427	0.478	0.517	0.353	0.380	0.552	0.555	1.415	0.879	3.561	1.473	3.446	1.460
	720	**0.433**	**0.412**	0.460	0.436	0.674	0.583	0.454	0.432	0.509	0.465	0.523	0.510	0.553	0.538	0.475	0.445	0.701	0.627	1.822	0.984	3.896	1.533	3.445	1.460
	Avg	**0.274**	**0.323**	0.308	0.346	0.399	0.426	0.314	0.352	0.344	0.372	0.381	0.404	0.388	0.433	0.341	0.372	0.534	0.547	1.415	0.871	3.658	1.489	3.581	1.487
$Weather$	96	0.172	0.263	0.175	0.230	0.184	0.242	**0.171**	**0.224**	0.207	0.253	0.229	0.309	0.227	0.299	0.215	0.252	0.218	0.295	0.230	0.285	0.497	0.497	0.406	0.435
	192	**0.224**	**0.271**	0.227	0.276	0.228	0.283	0.230	0.277	0.272	0.307	0.265	0.317	0.278	0.333	0.290	0.307	0.294	0.331	0.274	0.323	0.620	0.545	0.446	0.450
	336	0.282	**0.321**	0.286	0.322	**0.279**	0.322	0.294	0.326	0.313	0.328	0.353	0.392	0.351	0.393	0.353	0.348	0.359	0.398	0.318	0.355	0.649	0.547	0.465	0.459
	720	0.366	0.381	0.366	**0.379**	**0.364**	0.388	0.384	0.387	0.400	0.385	0.391	0.394	0.387	0.389	0.452	0.407	0.461	0.461	0.401	0.418	0.570	0.522	0.471	0.468
	Avg	**0.260**	0.309	0.263	**0.301**	0.263	0.308	0.269	0.303	0.298	0.318	0.309	0.353	0.310	0.353	0.327	0.328	0.333	0.371	0.305	0.345	0.584	0.527	0.447	0.453
$Electricity$	96	0.147	0.242	**0.143**	**0.241**	0.150	0.251	0.145	0.244	0.315	0.389	0.235	0.322	0.297	0.367	0.484	0.518	0.697	0.638	0.639	0.609	1.265	0.919	1.414	0.855
	192	**0.158**	**0.241**	0.159	0.255	0.163	0.263	0.163	0.260	0.318	0.396	0.247	0.341	0.308	0.375	0.501	0.531	0.718	0.648	0.772	0.678	1.298	0.939	1.240	0.919
	336	0.178	0.277	0.179	**0.274**	**0.175**	0.278	0.183	0.281	0.340	0.415	0.267	0.356	0.354	0.411	0.574	0.578	0.758	0.667	0.901	0.745	1.302	0.942	1.253	0.921
	720	0.224	0.312	0.233	0.323	**0.219**	**0.311**	0.233	0.323	0.635	0.613	0.318	0.394	0.426	0.466	0.952	0.786	1.028	0.788	1.200	0.871	1.259	0.919	1.249	0.921
	Avg	0.179	**0.268**	**0.178**	0.273	0.176	0.275	0.181	0.277	0.402	0.453	0.266	0.353	0.346	0.404	0.627	0.603	0.800	0.685	0.878	0.725	1.281	0.929	1.289	0.904
$Traffic$	96	0.414	0.291	0.419	0.298	0.427	0.304	**0.404**	**0.286**	0.854	0.492	0.670	0.421	0.795	0.481	1.468	0.821	1.643	0.855	1.157	0.636	1.557	0.821	1.586	0.841
	192	0.419	**0.291**	0.434	0.305	0.447	0.315	**0.412**	0.294	0.894	0.517	0.653	0.405	0.837	0.503	1.509	0.838	1.856	0.928	1.688	0.848	1.596	0.834	1.602	0.844
	336	**0.437**	0.314	0.449	0.313	0.478	0.333	0.439	**0.310**	0.853	0.471	0.707	0.445	0.867	0.523	1.602	0.860	2.080	0.999	1.826	0.903	1.621	0.841	1.668	0.868
	720	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
	Avg	0.423	0.298	0.434	0.305	0.450	0.317	**0.418**	**0.296**	0.867	0.493	0.676	0.423	0.833	0.502	1.526	0.839	1.859	0.927	1.557	0.795	1.591	0.832	1.618	0.851
1st Count		**21**		6		7		6		0		1		0		0		0		0		0		0	

forecasting tasks in terms of the total number of trainable parameters, GPU memory overhead, and training speed. Quantitatively, there is an **71.2%** trainable parameter reduction on average over four scenarios, leading to **23.1%** smaller memory consumption and **25.3%** faster training speed.

H Error Bars

All experiments have been conducted three times, and we present the standard deviations of our model and the runner-up model here. The comparisons between our method and the second-best method, PatchTST (Nie et al., 2023), on long-term forecasting tasks, are delineated in Tab. 19. In this table, the average MSE and MAE have been reported across four ETT datasets, complete with standard deviations. Furthermore, Tab. 20 contrasts the effectiveness of our method with that of the second-best method, N-HiTS (Challu et al., 2023a), employing varying M4 datasets for the comparison.

I Visualization

In this part, we visualize the forecasting results of TIME-LLM compared with the state-of-the-art and representative methods (e.g., GPT4TS (Zhou et al., 2023a), PatchTST (Nie et al., 2023), and Autoformer (Wu et al., 2021)) in various scenarios to demonstrate the superior performance of TIME-LLM.

In Fig. 7 and Fig. 8, the long-term (input-96-predict-96) and short-term (input-36-predict-36) forecasts of various approaches are compared with the ground truth. Here, TIME-LLM showcases forecasting accuracy that is notably superior compared to GPT4TS, PatchTST, and a classical Transformer-based method, Autoformer.

We also offer visual comparisons of the forecasting results in both few-shot and zero-shot scenarios, as depicted in Fig. 9 and Fig. 10. We adhere to the long-term (input-96-predict-96) forecasting setup